BIRD SPECIES OBSERVATION ANALYSIS

Ecological Data Analysis: Avian Species Monitoring & Conservation Insights

Executive Summary

This project analyzes a comprehensive dataset of bird observations across various national park plots. The objective was to clean the raw data, engineer a relational database for efficient querying, and build an interactive dashboard to provide park management with actionable insights on species populations, "At-Risk" conservation tracking, and optimal survey conditions.

Phase 1: Exploratory Data Analysis (EDA) with Python

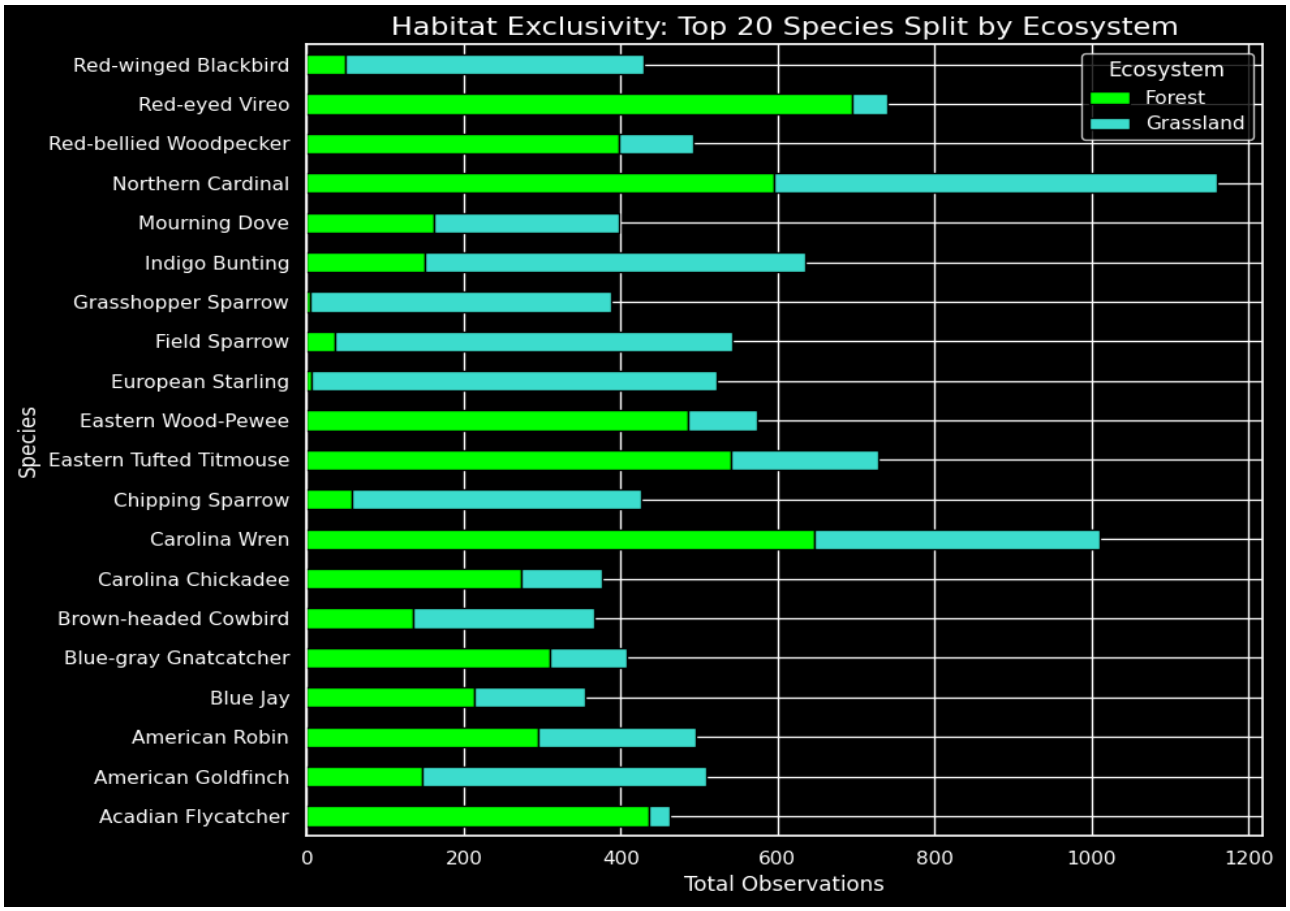
Objective & Planning:

The initial phase focused on understanding the shape of the raw data, identifying missing or inconsistent values, and uncovering baseline trends before loading the data into a formal database.

Procedures:

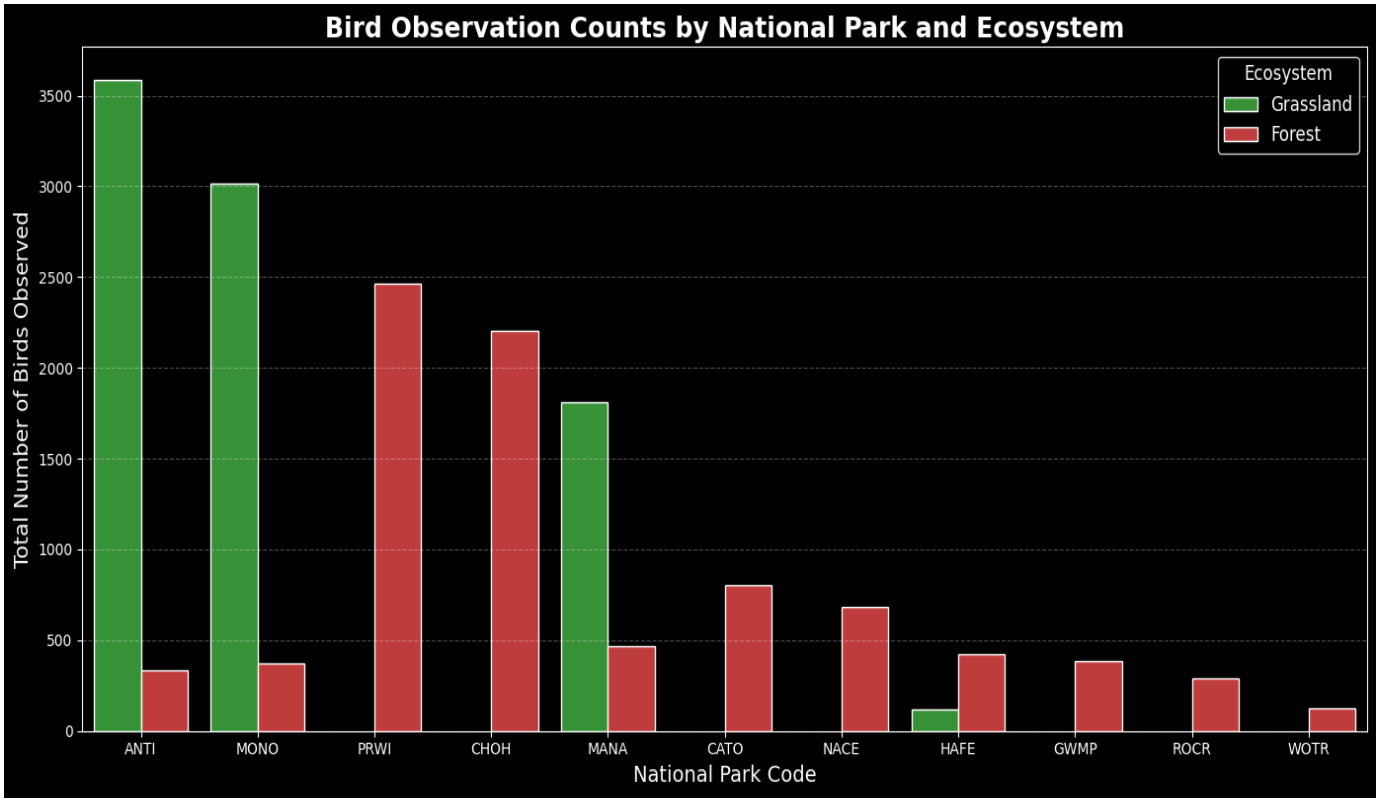
- ***Data Cleaning**: Handled missing values, standardized text columns, and verified data types using Pandas.*
- ***Descriptive Statistics**: Generated summary statistics for environmental factors (Temperature, Humidity) and observation counts.*

- ***Initial Visualization**: Utilized Matplotlib and Seaborn to plot the distribution of species across different ecosystems and analyze the methods used for bird identification.*



Key Insights:

- ***Detection Methods**: Visual sightings are remarkably rare. The vast majority of birds are identified via audio (Singing/Calling), indicating that park budgets should prioritize high-quality acoustic monitoring equipment over visual aids.*
- ***Ecosystem Dominance**: Certain environments (like Grasslands vs. Forests) show distinct variations in species volume, establishing the need for targeted, habitat-specific conservation strategies.*



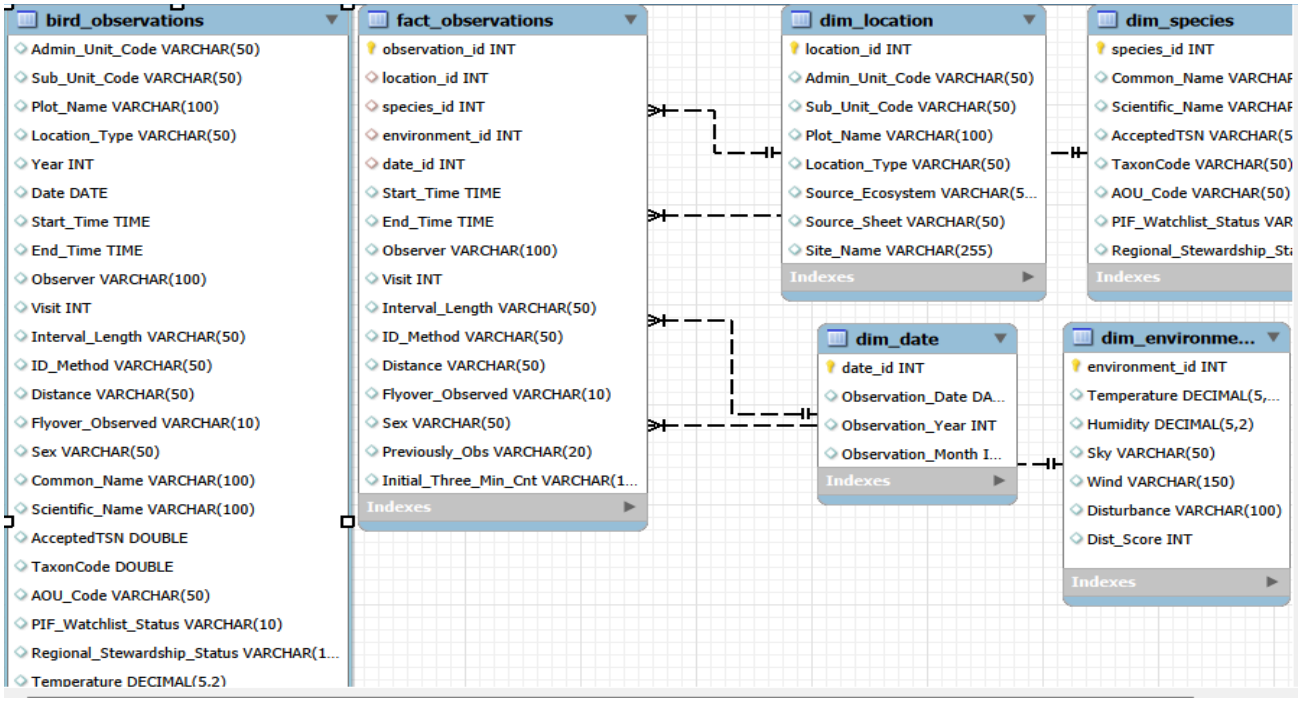
Phase 2: Data Engineering & Advanced Querying (SQL)

Objective & Planning:

To eliminate data redundancy and improve query performance, the flat CSV file was transformed into a normalized relational database. The goal was to build an architecture capable of supporting complex analytics.

Procedures:

- ***Database Architecture*** **:** *Designed and implemented a Star Schema consisting of one central Fact Table (`fact_observations`) and four surrounding Dimension Tables (`dim_location`, `dim_species`, `dim_environment`, `dim_date`).*
- ***Data Migration:*** *Used `INSERT INTO ... SELECT DISTINCT` statements to populate the dimension tables and map foreign keys.*
- ***Advanced Analytics:*** *Executed complex queries utilizing CTEs (Common Table Expressions) and Window Functions (`DENSE_RANK`, `LAG`, `SUM OVER`) to calculate running totals, moving averages, and percentile rankings.*



Key Insights:

- ***Protocol Integrity:*** *Self-joins revealed that there was zero observer overlap on the same plot on the same day, proving that strict ecological survey protocols were followed to prevent double-counting.*
- ***Conservation Tracking:*** *Successfully isolated "At-Risk" species (based on PIF Watchlist status) to track their unique discovery dates and cumulative sightings over the season.*

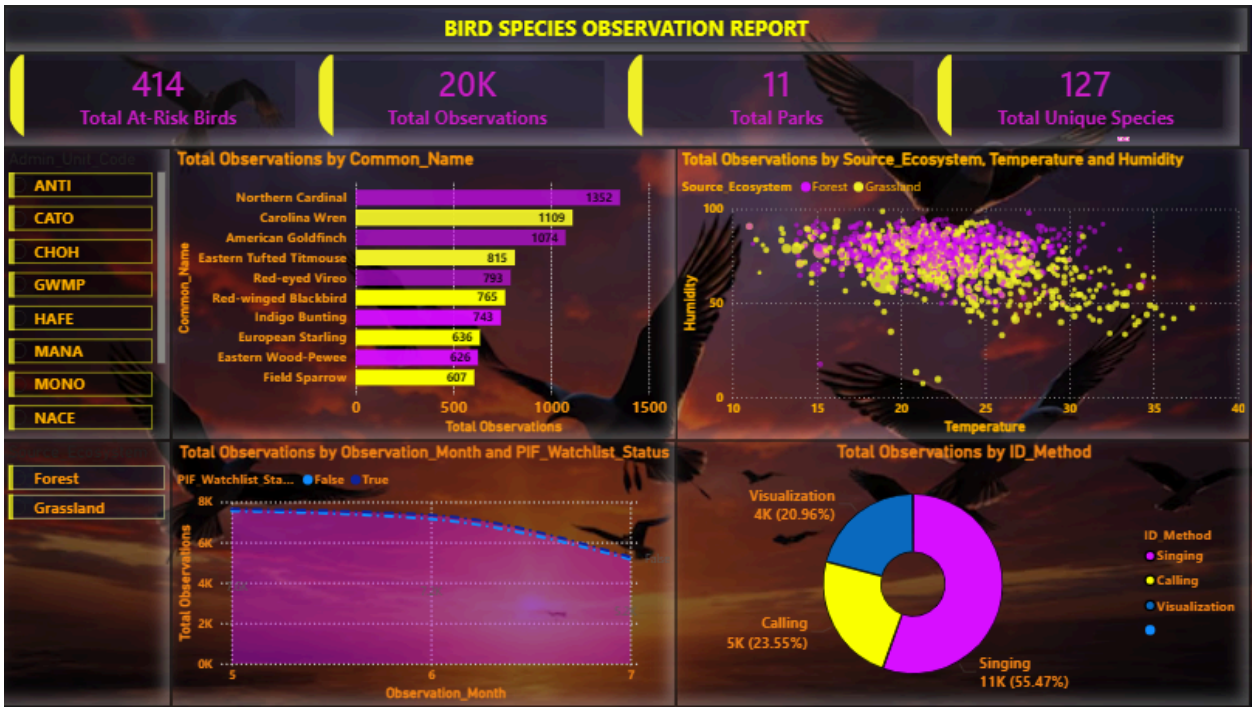
Phase 3: Interactive Dashboarding (Power BI)

Objective & Planning:

The final phase translated the engineered data into a dynamic, user-friendly interface. The goal was to create a "Night Mode" command center with high-contrast visuals, allowing stakeholders to filter data instantly without writing code.

Procedures:

- ***Data Modeling:*** *Connected Power BI directly to the MySQL database, establishing 1-to-Many relationships between the Fact and Dimension tables.*
- ***DAX Engineering:*** *Authored custom DAX measures (e.g., `CALCULATE`, `DISTINCTCOUNT`) to track Total Observations, At-Risk Populations, and Unique Parks Surveyed.*
- ***Visual Design:*** *Applied a custom dark-theme layout using a "Neon Nature" color palette (Cyan, Neon Green, Orange) to maximize contrast and readability. Implemented interactive slicers for Park Names and Ecosystems.*



Key Insights:

- ***Top Species Profiles:*** *The dashboard instantly identifies the top 10 most abundant birds, dynamically adjusting based on the selected park or habitat.*
- ***Seasonal Surges:*** *The "Observations Over Time" area chart highlights exact seasonal spikes, allowing park rangers to optimize staffing during peak migration or breeding windows.*
- ***The Weather "Sweet Spot":*** *A scatter plot mapping Temperature vs. Humidity reveals the exact environmental conditions that yield the highest volume of bird activity, enabling highly efficient scheduling for future surveys.*